Molecular Dynamics Simulations with Advanced Sampling to Identify Novel PLK1 Inhibitors

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Abstract

This proposal demonstrates the necessity of advanced computing resources in our research at the **Drug Design Lab at Gallaudet University** to identify polo-like kinase 1 (PLK1) via molecular dynamics (MD) simulations with advanced sampling. In addition, the study will establish the molecular mechanism of potential PLK1 inhibitors as anticancer drugs. PLK1 is a serine/threonine kinase with a crucial role in the cell cycle; it is essential to CDK1 phosphorylation, which later activates the mitotic machinery and triggers cell division[1]. PLK1 overexpression is a remarkable feature of numerous cancer types, making its inhibition a highly promising strategy for cancer therapy[2]. One of the several ways to target PLK1 is to use small molecule inhibitors that block the kinase activity of PLK1. *In vitro* and *in vivo*[3] studies have shown these inhibitors' effectiveness against cancer cells. PLK1 inhibitor design and development continue to gain more attention because they are potentially more effective and less toxic treatments than traditional cancer therapies[2,3]. The central goal of our research is to identify specific and selective PLK1 inhibitors by studying their binding profiles plus structural and dynamic properties through atomistic molecular dynamics (MD) simulations with advanced sampling. Specifically, we aim to use Metadynamics (MetaD) that accelerates the sampling of conformational events and estimates the free energy to reveal these inhibitors' binding and catalytic mechanisms when bound to PLK1. This approach will enable us to understand the PLK1-ligand complex that supports the structural activity relationship (SAR) of PLK1 inhibition. To determine the selectivity of the identified compounds, we will perform the same MetaD simulations for PLK2 and PLK3 ligand complexes. We intend to simulate about 150 small-molecule drug candidates identified through virtual screening. In our previous work, we established the binding site diversity of PLK1 with previously known and experimentally studied molecules with PLK1 inhibition potential. This work utilized supercomputing resources from ACCESS Discover and Accelerate resources, and publications describing it are currently under peer review in two ACS journals. Our MetaD simulations will dwell on findings from these studies and further explore the potential energy landscape more accurately than standard MD and subsequently calculate the binding free energy of the different ligand–PLK1 complexes. However, the MetaD for 300 ligand-protein complexes will require a considerable supercomputing time, which is being sought through this proposal. In addition to molecular simulations, this work will involve complementary fluorescence polarization experiments here in our lab to measure the binding strength of each ligand to PLK1 and Circular Dichroism experiments to decipher structural changes to PLK1 upon ligand binding. Therefore, we can directly validate and further develop our computational work, enabling a massive outcome.

Keywords: Advanced sampling, Metadynamics, Computational drug discovery, PLK1 inhibition, Anticancer drugs, Breast cancer, Cancer metastasis.

1. Introduction

[Our research is funded by National Institutes of Health grant 1R15GM148942-01, which provides stipend funds for our research group members but not supercomputing time or instrument purchase.]

[Our previous ACCESS Discover and Accelerate allocations, although limited, resulted in significant research developments. More information on the most recent research and the two articles under review are provided in the subsequent section. Our allocations were utilized by a postdoctoral researcher, a collaborator faculty, and two students.]

PLK1 plays a crucial role in regulating p53-mediated damaged DNA repair and mitosis (**Figure 1**), also indicative of cellular proliferation with prognosis prospects in certain types of cancers[4–6]. Studies have shown a connection between a high level of PLK1 expression and a poor prognosis — diminished PLK1 expression

may lead to apoptosis and hinder several cancer cells' growth[7–9]. Targeting and inhibiting PLK1 is a promising therapeutic approach for various types of cancer, including metastatic breast cancer and multidrug-resistant tumors[3,10–14]. As part of an awarded grant by Atomwise company, 46 small molecules were identified as plausible PLK1 inhibitors for our target protein PLK1. The compounds were purchased and shipped to Gallaudet for fluorescence polarization experiments to measure their binding to PLK1. To complement and further guide this experimental work, we will study the interaction of these small molecules with the target protein PLK1 at the atomic level through conventional and advanced sampling MD simulations. Also, we aim to explore the interaction mechanism and dynamical changes of these 46 compounds at the molecular level through MetaD. This MetaD sampling approach will allow us to sample rare events in PLK1–3 and small molecule interactions and analyze the protein-inhibitor relationship. This theoretical approach will enable us to establish the properties behind favorable inhibitor binding and uncover conformational states that would seldom be explorable with classical MD.

Study Goals: The overarching goal of our study is to understand the detailed free energy surface during the binding of identified novel nonpeptide-based selective inhibitors for the polo-box domain (PBD) of polo-like kinase 1 (PLK1). Specifically, MetaD simulations will be performed for two sets of molecules. First, the previously studied inhibitors with known experimental binding sites will be studied. Then, the newly identified possible PLK1-PBD molecules will be performed.

Specific Aim 1: Metadynamics simulations of 46 newly identified potent PLK1-PBD inhibitors.

Specific Aim 2: Metadynamics simulations of >100 identified PBD inhibitors from natural compound databases.

Significance

Given its prime role in mitotic events, PLK1 is a marker for cellular proliferation with prognosis potential in some types of cancers[4–6]. A high level of PLK1 expression is associated with a poor prognosis[7–9]. Studies have shown that diminishing PLK1 expression can induce apoptosis and inhibit growth in many cancer cells[7,15] but showed an insignificant effect on normal cells in mouse models[16]. On the other hand, PLK2 and PLK3 are signatory tumor suppressors[17,18]. Therefore, selective and specific inhibition of PLK1 is a promising therapeutic strategy for a range of cancer types, which has been suggested, particularly for metastatic breast cancer and for tumors with acquired resistance to earlier targeted therapies and (or) endocrine therapies[3,10–14].

PLK1 is a conserved serine-threonine protein kinase required for a broad regulatory spectrum of cell cycle and mitosis events. It is an early trigger for G2/M transition and regulates diverse mitotic events during M-phase progression. PLK1 also acts as an indirect regulator in DNA replication and metabolism, such as glycolysis through PTEN (phosphatase and tensin homolog)[8,19]. PLK1 has two distinct druggable targets: The highly conserved N-terminal kinase catalytic domain (KCD) and the C-terminal noncatalytic region with the polo-box domain (PBD) with the polo-box units joined by a 50-residue linker. The latter is unique to PLK1–3 out of the family of five PLKs identified in mammals, and it is responsible for its subcellular localization and mitotic functions through binding to phosphorylated substrates[1].

The medical sector has witnessed significant progress in generating KCD inhibitors, which yielded several ATP-competitive analogs currently in use (volasertib) or clinical trials (such as rigosertib). However, due to a high degree of conservation among kinases, KCD poses selectivity problems that exhibit dose-limiting toxicities[9,20,21]. Therefore, PBD has emerged as an alternative target for developing a new class of PLK1 inhibitors. A substantial binding profile, high specificity level, and low toxicity make PBD inhibitors attractive as an anticancer drug target that could be used alone or in combination with available KCD inhibitors[3,7,9,14,21–27]. For these reasons, our study targets the PBD of PLK1.

Expected Outcomes

Detailed modeling of PLK1–3 and potent inhibitors using MetaD will facilitate an improved understanding of these inhibitors and selectivity for PLK1 at the atomistic level. The Free energy pathway would show the most promising and energetically favorable protein–inhibitor complex. Other analyses include calculating

the binding free energy and its components, conformational study, and interaction profile to unravel the most crucial energy term behind optimal PLK1–inhibitor binding and unbinding episodes, large-scale motions, ligand orientation, and residue of importance. Analysis of the H-bonding network will provide insight into the protein's structural integrity, PLK–inhibitor stability, electrostatic properties, and binding modes. The interrelationship of these outcomes would improve the accuracy of the study. Our analyses will allow us to uncouple the structural properties of the three PLK proteins–ligands, receptors, and inhibitors only.

Overall, this investigation will provide an atomistic understanding of interactions, including electrostatic and van der Waals in each ligand investigated and PLK1, PLK2, and PLK3 protein structures. The currently highlighted implications are threefold. (1) This exploration will show the crucial interactions involved in inhibitor binding strongly compared to weak binding. (2) We will compare the interactions across receptors to reveal the selectivity of each ligand. (3) The study will also unveil interactions across different ligands to decipher their inhibitory strength in comparison with the experiment towards validating our MetaD protocol. The major challenge in these simulations is to obtain the correct parameters for the small molecules. In contrast, the great opportunity is that experimental relative binding information is available to validate our method. If the results are inconsistent with the published data, we plan to compare the current parameters with CGenFF[28] parameters and make necessary changes based on insights gained from the results.

Potential Impact of the Project on Broader Fields: The proposed project has a multifaceted approach involving multiscale simulations, fluorescence polarization experiments, and circular dichroism (CD) spectroscopy experiments. The study of PLK1 targeting cancer therapy will significantly impact the fields of oncology, medicine, and drug design. Our study will lead to new and promising cancer therapies development, broadening the insights into the molecular mechanism of PLK1 in cancer pathology. The identified active small molecules targeting PLK1 will contribute to developing novel inhibitor designs selective to block the kinase activity of PLK1. The laboratory experiments will enable us to predict the effectiveness of the inhibitors in killing cancer cells and inhibiting PLK1. Besides, this research will benefit combinatorial cancer therapeutics. For instance, PLK1 inhibitors could enhance the efficacy of chemotherapy drugs, radiation therapy, and immunotherapy[3]. This discovery suggests that PLK1 targeting may be a particularly effective treatment for cancers resistant to other therapeutics. The study of PLK1 inhibition would provide more insights into cancer prognosis since PLK1 plays a regulatory role in cell cycle progression, DNA damage repair, and apoptosis[4–6]. Therefore, our study would enhance new target identification for cancer therapy while facilitating new biomarkers development that could potentially track cancer progression and response to treatment assessment.

Using MetaD protocol to study the PLK1–inhibitor systems will contribute to several topics in biophysical sciences like mechanistic study, enzymology, and method optimization in protein modeling. The sampling approach will offer an improved perspective on free energy, binding energy, and thermodynamics components. The parameterization protocol for MD and MetaD protocol for ligand-protein systems will apply to other PLK families or related protein structures. Such knowledge will contribute to advanced computational and theoretical methodologies in protein modeling. Tuning the MetaD sampling strategy and identifying the correct force field parameters for the nonpeptide small molecules would interest audiences, including computational chemists, method developers, and drug designers. Specifically, this study is significant with an excellent potential to guide near future *in-vivo* studies or further clinical considerations of PLK1-PBD inhibitors.

Research and Educational Objectives: The objective is to perform MD simulations and estimate free energy by adding bias potential to our PLK1–potent inhibitor systems. MD simulations with NAMD will enable us to generate various conformational ensembles of PLK1–inhibitors. The comparative simulations to explore selectivity will involve all relevant PLK family of kinases (PLK1–3). Besides exploring the free energy landscape of these protein-ligand systems, the MetaD approach will enable us to unravel metastable states, including possible transition states (TSs), that would be difficult to obtain using experimental methods. Our group has three faculty members, one postdoctoral associate, postbaccalaureate research assistants, and undergraduate and high school students. We employ a multiscale simulation approach, which includes standard MD, evaluation of drug-likeness properties, free energy calculations utilizing fast webservers, and

MD with advanced sampling (which will be enabled through the approval of this proposal). With this diversity of methods and diverse academic levels of team members (from faculty to high school), we employ a bilateral mentoring approach. In this approach, we pair experienced participants with newcomers, providing our researchers with opportunities to mentor. While doing so, we develop bilingual (American Sign Language and English) tutorials for each completed simulation. The supercomputing resources obtained through this proposal will be vital to advancing our research and mentoring goals with deaf and hard-of-hearing (D/HH) students at Gallaudet and nationwide.

Justification

Project Team Description: Gallaudet University is a private federally chartered university in Washington, D.C., for the education of the deaf and hard of hearing. The Drug Design Lab of the School of Science, Technology, Accessibility, Mathematics, and Public Health (STAMP) at Gallaudet is led by Dr. Kucukkal, who has pioneered providing D/HH students opportunities in research since joining Gallaudet in 2016. She has so far mentored 38 deaf and hard-of-hearing students. Our Research Team consists of two deaf co-investigators, Dr. Adebowale Ogunjirin and Dr. Daniel Lundberg, both of whom have been actively involved in student mentoring for more than ten years and have been excellent deaf scientist role models for our students. In addition, we have a postdoctoral research associate with a proven record of publications in molecular modeling and has authored three manuscripts since her start at Gallaudet in June 2023. We also have five undergraduate students, two postbaccalaureate researchers who are recent graduates from Gallaudet, and several high school students. The Drug Design Lab at Gallaudet University also works with hearing students through Quest Student Research Institute (QUESTSRI) and more D/HH students nationwide through Hillman Academy at the University of Pittsburgh and James Madison University REU program. We also collaborate with Dr. Shanen Sherrer at the St. Mary's College of Maryland and Dr. Nate Wright of James Madison University. Over the years, we have been using various open-source software packages for computational drug design research projects and student training. Lately, Dr. Kucukkal and Team acquired a $368,671 grant from the National Institute of General Medical Sciences (NIGMS) of the National Institutes of Health to research nonpeptide potent inhibitors of PLK1 as cancer therapy.

Justification of Requested Resources: Our prior work included standard MD simulations with the MMPBSA approach to decipher the binding mechanism of PLK1-PBD with previously identified small molecules. We also examined the drug-likeness properties of 46 small molecules of PLK1 inhibitors identified by Atomwise[29]. The software packages and online tools utilized include VMD[30], Avogadro[31], CB-Dock2[32], FitDock[33], SwissADME[34], NAMD[35], ProteinTools[36], PRODIGY[37], and CSM-lig[38] for various simulations. In the next phase, we plan on performing MetaD simulations that will dwell on previous knowledge obtained about the binding mechanism and binding diversity of PLK1-PBD. Our experience utilizing ACCESS resources previously and pilot studies for MetaD justifies the need for computational resources beyond those provided by the Accelerate program. Our local computing capability, as described below, is insufficient for any advanced sampling.

Local Computing Resources: Our current hardware infrastructure includes 2 NVIDIA workstation laptops and a 16-core CPU supercomputer. This hardware is insufficient to support our operations for future steps in the current project. Most recently, Dr. Kucukkal was awarded the President's Research Excellence grant from Gallaudet University to purchase 4 GPU-based workstations to execute research aims and train students. These resources will also be utilized for the current study. However, the priority for those resources would be student training, which will include standard MD simulations.

Previous Findings Utilizing ACCESS Discover and Accelerate Resources

We summarize our recent research studies on analyzing the fundamental interactions between selected small molecules and the PLK1-PBD through classical MD (cMD) simulations[39] and druggability screening of newly identified potent PBD inhibitors[40].

MD Simulations of Experimentally Studied PLK1-PBD Inhibitors: Utilizing the previous ACCESS Accelerate allocation, we have systemically investigated the binding of eight previously studied small molecules and a phospho-peptide binding to PBD1 at the atomic level as well as the unbinding of phospho-peptide through adaptive steered MD simulations. After blind docking these molecules, at least 5 primary binding sites (BS1, BS4, BS5, BS6, and BS7) were identified, which are consistent with the results of our comprehensive binding site screening of 50 available PBD crystal structures from the protein data bank (PDB)[41]. Labeled as BS1-7, we closely investigated possible binding sites which are shown in Figure 1. BS4 stood out for its depth and stability, but compound (–)–9* and (+)–9* did not favor BS4 due to their bulkier structures. On the other hand, blind docking did not confirm BS1-2 (phospho-peptide and (BS1) Tyr-rich (BS2) pocket) and BS3 (W-F pocket) as binding sites, likely because it relies on PBD1's flexibility to accommodate inhibitors. While the lowest docking score was observed for Thymoquinone (TQ)[42] binding at the canonical phospho-substrate site, the highest ligand binding affinity was observed with Allopole-A[43] at the W-F cleft of the PLK1 polo-box domain (**Figure 1**), aligning with the experimentally predicted inhibitory region. Compounds (–)–9* and (+)–9*[44] displayed high docking scores (–6.3 to –7.5 kcal/mol), and Poloxin derivatives[42,45] exhibited similar scores per binding site.

Further analysis of protein flexibility from our comprehensive all-atom cMD simulations (total simulation time of 66 microseconds) revealed significant fluctuations in various loop regions, with the apo state displaying the highest flexibility and the phospho-peptide-bound state showing the lowest[39]. Notably, the L2 loop connecting the two polo boxes of PBD1 was vital in maintaining binding pocket stability. The clustering analysis of trajectories identified three distinct conformational states (cS1, cS2, and cS3) across all simulations, indicating that the ligands and phospho-peptide preferred pre-existing PBD1 conformations (**Table 1**) for binding. These states were found in alignment with available experimental X-ray crystallography structures. The probability analysis showed varying degrees of conformational selection among different inhibitors, highlighting their binding preferences. This diversity in binding preferences underscores the importance of understanding PBD1 pockets' symmetries and the specific configurations favored by each inhibitor binding[39].

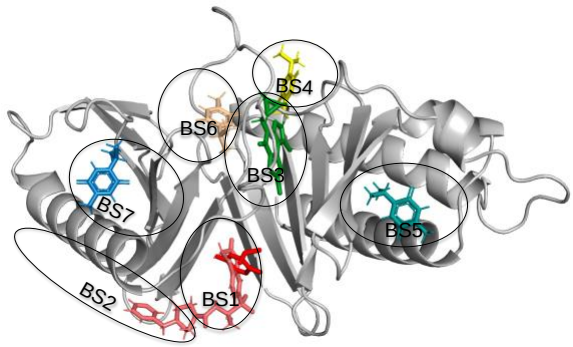

Figure 1. Experimentally identified binding sites (BS, 1 to 3) in PBD, blind docking revealed inhibitor binding to five possible pockets, including BS1 (phospho-peptide pocket).

Table 1. Percent (%) probability of the three PBD1 conformational states (cS) in apo, phospho-peptide (PP), and selected inhibitor (PLX indicates Poloxin).

cS	Apo	PP	TQ	PLX	PLX-2	PLX-22	PLX-23	(–)–9*	(+)–9*	Allopole-A
cS1	58	16	73	44	30	54	26	74	73	37
cS2	17	34	7	42	63	12	8	21	13	52
cS3	25	50	20	14	7	34	66	5	14	11

Quantitative aspects of binding energetics through binding energy calculations provided valuable insights into the thermodynamics of interactions with PBD1[39]. The binding free energy values obtained from MM-PBSA analysis (**Figure 2**) generally agree with the experimental inhibition profile, except (+)–9* which is excluded in the figure. We suggest that using MetaD will significantly improve the accuracy of binding energy calculations. MetaD will be advantageous in unveiling hidden conformational ensembles and the entire repertoire of shapes and movements that a protein-ligand complex can adopt. This approach is critical for understanding alternative binding modes, allosteric effects, and protein flexibility[46]. Additionally, upon observing the stability of certain inhibitors at the various binding sites suggesting allosteric binding effects, we further examined them through adaptive steered MD simulations[39]. Before moving on to ASMD, electric field analysis reveals potential catalytic mechanisms, such as N—C=O hydrolysis in Allopole-A and (–)–9*. Specific residues like Arg500 and Glu504 in BS4 proved crucial. Poloxin-PBD1 BS5 shows 16 crucial residues, including Asp554 and Arg560. Compound (–)–9* in BS7 involves Lys413, Lys492, and Arg557.

These compounds may impact bioactivity, requiring further study. While the adaptive steered MD simulations confirm the allosteric potential of Allopole-A, poloxin was found to have negligible allosteric modulation on PBD1, mirroring a non-competitive inhibition[39]. On the other hand, the compound (–)–9* was well-suited as a competitive inhibitor, as proposed in the literature[44].

One research question that has not been addressed in detail is the role of water in PLK1-PBD dynamics and ligand interactions. Using MetaD to probe water's crucial role after explicit solvation would provide insights into their dynamic interplay with the protein–ligand complex[47]. Also, advanced MD captures rare events like conformational transitions better than cMD, providing valuable information about binding mechanisms and ligand release pathways. Improving the sampling approach could facilitate better drug design by understanding the dynamic interaction landscape and tailoring ligands to fit more tightly and selectively into their target proteins. Although enhanced sampling is resource-expensive, achieving an exhaustive approach to probe some queries around PLK1 versus PLK2&3 proteins will pave the way for more effective, selective, and targeted therapeutics for cancer.

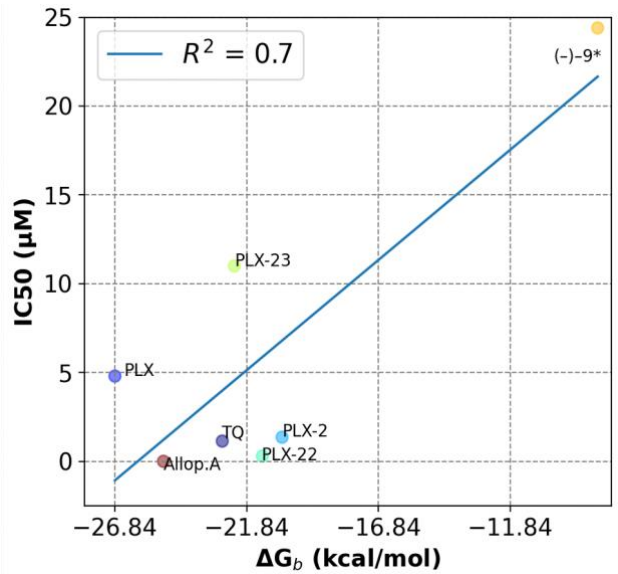

Figure 2. Correlating experimental IC_{50} (in μM) values of selected ligands, including Thymoquinone (TQ), Poloxin (PLX), and Allop.A (Allopole-A) with calculated energy values for PBD1 inhibition. The most favorable ΔG_{bind} (in kcal/mol) per inhibitor (excluding (+)–9*) correlates significantly with IC_{50}.

Physicochemical, drug-likeness, and ADMET-S properties of small molecules: 46 potential PLK1-PBD inhibitors were evaluated by predicting their physicochemical properties, drug-likeness, ADMET (absorption, distribution, metabolism, excretion, toxicity) profiles, and stability[40]. Despite their ease of use, processing the complex data from software like ADMETlab with about eighty-eight metrics to rank the molecules was found to be the challenging part of the study for the students. The BOILED-egg approach and radar charts proved vital in analyzing the data. The results reveal several promising candidates that exhibit desirable properties across multiple criteria. Compound **30** is the most promising PLK1-PBD candidate drug, demonstrating favorable attributes in physicochemical properties, drug rule, and pharmacokinetics outcomes. Several other compounds also show potential and warrant further investigation[40]. Overall, this research advances our understanding of evaluating small molecule properties while contributing to research skills development and learning opportunities for students and researchers[40].

The cMD study[39] and other analytic approaches facilitated the crucial interactions and structural components mastery of the protein required for optimum binding to the selected small molecules at various binding pockets. This knowledge enabled us to suggest the variables for the MetaD simulation setup. We are also screening natural product databases to uncover natural compounds that would inhibit PLK1-PBD. Together with the Atomwise 46 small molecule dataset, all identified top promising novel PLK1-PBD inhibitor candidates in our current anticancer research will require MetaD simulations to establish reaction mechanism and free energy surface.

Computational Methods

Metadynamics

Metadynamics is a widely recognized sampling technique utilized in MD simulations. This method reconstructs the free-energy surface by carefully choosing collective variables (CVs)[48–50]. MetaD is a valuable tool employed to improve the sampling of rare events and model the free energy landscape of molecular systems by introducing a biased potential energy surface, which enables the exploration of energy barriers and the prediction of reaction rates[46]. The fundamental idea behind MetaD is to add a time-dependent bias potential to the original potential energy surface (PES)[51]. This bias potential construction

arises from depositing Gaussian-shaped hills along the system's collective variables[52]. These selected CVs often represent degrees of freedom like bond lengths, angles, or torsional angles[48–50]. Metadynamics encourages the system to explore regions of the phase space that would otherwise be unlikely to be visited by gradually increasing the height and width of the Gaussian hills[53]. As the simulation progresses, the bias potential accumulates in regions of low energy, effectively flattening the free energy surface, allowing free energy minima identification and thermodynamic properties calculation[49,53,54]. The accumulated bias potential also facilitates the underlying free energy surface reconstruction, which provides valuable insights into the system's behavior[49,50].

MetaD has successfully been applied to several scientific problems, including protein folding, ligand binding, and chemical reactions[49]. It is applicable in studying rare events like conformational changes or chemical reactions with high activation by accelerating the sampling of these events to facilitate kinetics and thermodynamics characterization[52,55,56]. The sampling approach is advantageous over other advanced sampling approaches because it does not require prior knowledge of the detailed energy pathway of the system[57]. MetaD first explores the low free-energy regions irrespective of the selected CVs to estimate dynamical properties like rate constants and mechanisms[56]. It allows exploring new reaction pathways as the system tends to escape the minima, passing through the lowest free-energy saddle point[49]. MetaD is self-regulating, with the potential to identify and adapt to convergence issues within the molecular system[55,58].

While not inherently a drawback, note that MetaD necessitates several parameters compared to alternative methods like umbrella sampling[54,55,58,59]. Despite its effectiveness, MetaD exhibits certain limitations. The choice of collective variables is crucial to the accuracy and efficiency of the method—it requires identifying a preliminary set of CVs (**s**) as a function of the system coordinates (**x**) to analyze the activated process in question[60]. Inappropriate or poorly chosen collective variables may lead to biased results or inefficient sampling. Hence, careful consideration and validation are necessary when selecting these variables. Our initial step involves utilizing MetaD simulations on PLK1–ligand complexes with established experimental binding profiles. This will help us ascertain the optimal choice of collective variables (CVs) for our subsequent simulations.

Principles of MetaD Simulations

The Gaussian potentials added during the MetaD simulations would discourage the system from revisiting previously explored states and push it towards exploring new regions of the PES[53]. The more the added Gaussian potentials, the more the system's energy landscape becomes progressively flattened, allowing for a more thorough exploration of phase space[48]. The height and width of the Gaussian potentials are relevant parameters in Metadynamics[51]. The height determines the strength of the biasing potential, while the width controls the extent of the explored regions[60]. Finding the optimal values for these parameters requires careful tuning to ensure an accurate system's free energy landscape representation[58]. Researchers often employ a well-tempered variant of Metadynamics to prevent the system from getting trapped in local minima[54]. In this variant, the implemented temperature replica exchange scheme allows the system to escape from shallow energy wells and explore different regions of the PES more efficiently. Metadynamics simulations are executable in various software packages, such as PLUMED[61], AMBER[62], GROMACS[63], and NAMD[35], which provide the necessary tools for setting up and analyzing the simulations. These software packages have options that enable users to select the CVs for customized simulation parameters that suit the user's specific needs.

Guiding Experimental Information

Previous studies identified several PLK1 ligands, including Poloxin[42,64], Thymoquinone (TQ)[65], and T521[1], with detailed interaction network profiling. Experimental analysis of human PLK1-PBD co-crystallized with TQ indicates Ser412, Lys413, Trp414, Leu491, Lys492, Ala493, Gly494, Asn533, His538, Lys540, and Arg557 contribution to the inhibitor binding and stability (**Figure 3**) via electrostatic and van der Waals forces[65]. Although these residues' interaction with TQ and other selected molecules at the phospho-epitope site agrees with our docking outcome, cMD studies and further refinement suggest otherwise. For instance,

seven potent PBD inhibitors displayed favorable binding free energies in the phospho-peptide pocket but with less stability[39]. This outcome is consistent with previous findings suggesting that several potent PBD inhibitors are protein alkylators with nonspecific cytotoxic effects[23,66]. Besides these selected inhibitors in our cMD studies[39], other potent compounds include T521[1], specific to inhibit the PBD of PLK1 via a covalent binding mechanism. The identified residues involved in T521 covalent binding to PLK1 are Lys209, Lys388, and Lys574, which are not within the conventional phospho-peptide binding pocket[1]. Other well-studied nonpeptide inhibitors of PLK analogs (PLK1–3) binding at the PBD site are Poloxipan[67], Purpurogallin[68], and Green Tea Catechins[69]. Crystallographic studies of human PLK1-PBD and its peptide complex showed that it makes short antiparallel β-sheet contacts via residues Trp414, Leu490, His538, and Lys540[70]. However, many of these inhibitors, including T521, have been tagged non-specific [23,66,71].

Experimental atomistic information on peptides and phospho-peptide derivatives binding at the PLK1-PBD is available in the PDB from several investigations. The co-crystallization of nonpeptide inhibitors attached to the PBD of PLK1 has appeared in a few studies[65,72,73], whereby TQ (**Figure 3**) binds preferentially at the phospho-mimetic binding pocket[65]. Other potent nonpeptide or phospho-based PBD inhibitors of PLK1 co-crystals include α-Bromo-3-Iodotoluene (5NJE[72]), Polotyrin (5NN2[72]), and JES107 (5NEI[73]) interaction on the PBD's αB helix surface through residues Val415, Tyr417, Tyr421, Leu478, Tyr481, Phe482, and Tyr485. These residues lined adjacent to the phospho-peptide binding groove with implications to regulate substrate recognition[72,74]. Although our target for the small nonpeptide inhibitors is the phospho-substrate pocket, we do not intend to rule out their probability of attaching to the PLK1-PBD Tyr pocket since we will be using a blind docking approach to couple these new inhibitors with the protein. Identifying the most crucial interactions and substructural moieties critical to binding existing molecules to PLK1-PBD will facilitate accurate CV selection for our identified novel inhibitors study using MetaD.

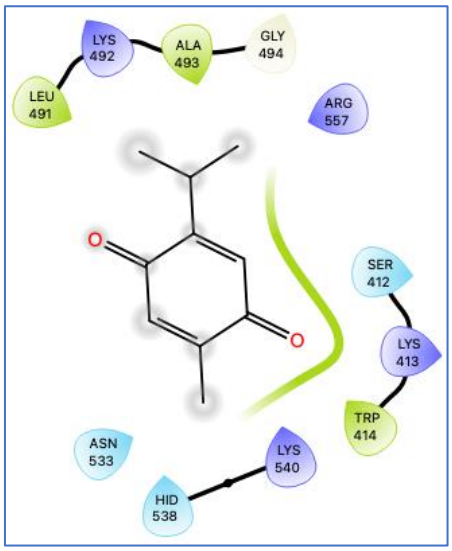

Figure 3. The interaction of TQ at the phospho-peptide binding site of PLK1-PBD from crystallographic information.

Computational Procedure in Our Application of MetaD

The first step of this research is a trial assessment and protocol design using existing potent PLK1-PBD inhibitors from experiments. Crystallographic studies of human PLK1-PBD and its peptide complex showed that it makes short antiparallel β-sheet contacts via residues Trp414, Leu490, His538, and Lys540[70]. Tyr417 and Tyr481 are among the most crucial residues in the Tyr pocket[72,74], while Ile406, Pro407, Trp410, and Phe559 are the essential W-F pocket[43] residues. Based on our cMD results and experimental information, the selected collective variables are the center of mass (COM) inter-binding site (BS) residues Trp414/Tyr417–His538/Lys540; Ile406/Pro407–Trp410/Phe559, and Asp554/Arg560–Glu575/Arg579 for (–)–9*; Allopole-A, and Poloxin binding to respective BS1-2; BS3; and BS5 pockets of PLK1-PBD. Therefore, the metadynamics script will look like the one herein for the three experimental PBD–ligand systems. Compound (–)–9*, Poloxin, and Allopole-A are excellent candidates to simulate at the atomistic level using MetaD simulations because available experimental information, including IC_{50} values, will facilitate validation and fine-tuned modeling. They also represent typical competitive, non-competitive, versus allosteric binding models. We expect a sufficient Gaussian sampling by setting our bias potential using the center of mass distances of

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critical binding residues. Still, if the experiment looks inaccurate, we will harness results from our adaptive steered MD (ASMD) and electric field simulation to propose another variable. This flexibility will enable us to explore the most critical collective variable and select the more accurate model for our newly identified PLK1-PBD potential inhibitors.

In a test-running study with compound (–)–9* at BS1-2 on the graphical processing unit, GPU-enabled AMBER20[62] using the Nonequilibrium Free Energy (NFE) toolkit, the obtained free energy path mimics a Gaussian distribution (**Figure 4**). The NFE was built to optimize metadynamics [48,75] for free energy calculations in cMD. The NFE toolkit provides the Adaptively Biased Molecular Dynamics (ABMD) method[76,77] that excels in its favorable scaling and short simulation times, requiring only a minimal set of control parameters. It calculates the potential of mean force (free energy landscape) as a function of user-defined CVs, enabling detailed analysis of molecular behavior at the atomic level. Using two groups of residues Trp414/Tyr417 and His538/Lys540 COM distance, the input *cv* file depicts the sampled residue atoms for the two groups separated by **0** at a maximum distance of 20 Å. The MetaD simulation uses a conformational flooding approach[48] involving an additional potential energy term that induces the system to search new regions of phase space. Adjusting parameters like temperature, simulation time, and resolution might considerably lower the energy barrier of this system.

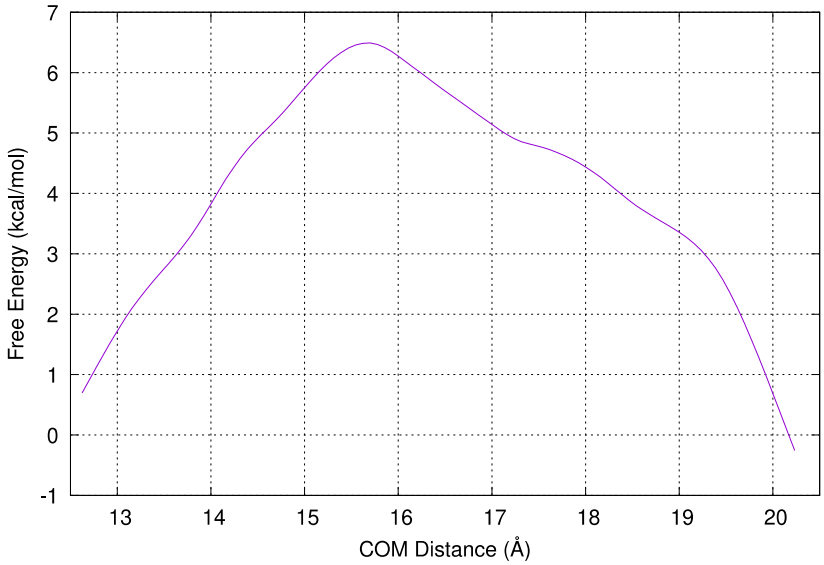

Figure 4. The resultant free energy plot for PLK1-PBD sampling when bound to (–)–9* using metadynamics in 1 ns production run. The CVs are the center of mass (COM) distance of crucial residues from the phospho-peptide–Tyr pocket.

The second stage is the blind docking of each molecule into each PLK1-PBD conformation using CB-Dock2[32] online software programs. The top five energetically favorable poses will become the starting models for the MetaD simulation. We have simulated the PBD structure (4HCO)[65] to obtain independent low conformational states starting protein ensembles using the Rosetta Backrub application[78], generating five independent starting protein structures.

The third step is protein and ligand preparation for simulations. With the top five ligands docked poses, the total protein–inhibitor states per binding pose for the MetaD simulations is five with at least 200 ns production. This setup will potentially result in 25 starting models per unique PBD–inhibitor, totaling 5 μs × 46 ABMD production. The selected force field for protein is the FF14SB[79] AMBER force field while parameterizing the ligand with the Generalized Amber Force Field (GAFF) obtainable from the AM1-BCC[80] Antechamber module. We will solvate each inhibitor state and apo PBD1 using the TIP3P water model in a 10 Å box edge and neutralize with 0.1 M sodium and chloride ions.

The fourth stage is simulations using AMBER20-GPU[62] on the Anvil infrastructure (anvil/architecture) with the CUDA version of the Particle mesh Ewald Molecular Dynamics (*pmemd.cuda*)[81] algorithm in 2 fs timescale integration. Other specifications include applying the SHAKE algorithm to constraint hydrogen atoms and Particle mesh Ewald for long-range electrostatic interactions at non-bonded interactions cutoff of 10 Å. The formalism for temperature and pressure controls are the Langevin thermostat and Monte Carlo barostat, respectively. Each protein-ligand setup proceeds in the *tleap* module of AMBER. To prevent distortion because of the GPU's rigorous approach to approximation, we will minimize, heat, and equilibrate each system gradually (pre-MD simulation stages). System minimization will run at 2500 steepest descent each for the restrained and non-restrained approach. Heating of the solute-solvent proceeded slowly from

0 to 300 K for 0.5 ns while applying 10 kcal·mol^{-1}·Å^{-2} force on the solute. The equilibration stage entails a 5 ns run before production in the NPT ensemble for at least 200 ns per replicate and switching on ABMD *FLOODING* mode[48].

The fifth step involves using the PLK1-PBD docking and simulation steps for PLK2 and PLK3 interaction with the 46 PLK1 molecules at the same conditions to predict selectivity. Since the PLK3 structure is unavailable, it will be obtained through AlphaFold[82] or homology modeling using SwissMODEL[83].

The sixth step is protocol adoption for other potential PLK1-PBD candidates identified from screening various natural compound databases. We have obtained about 70 molecules from the COCONUT[84] database and continue to explore other databases to identify promising PLK1-PBD molecules.

The final stage is post-simulation analysis with various metrics to compare the calculated results among the PLK proteins. The biased energy generated from the ABMD approach will be refined to predict the Free energy pathway for each system. Our simulations will determine their binding profile to PLK1-PBD, providing comparative results with experimental values from our lab. The study will show their binding to PLK2 and PLK3, shedding light on their potential for selectivity. Achieving selectivity for PLK1 is expected to be challenging but possible, as the overall sequence identity with the PLK1-PBD is 39% for PLK2 and PLK3. Although the similarity is higher in pertinent contact residues, there is a suggestion that selectivity may be achievable by targeting the Thr477/Leu478 pair at the end of the cryptic hydrophobic pocket, equivalent to Thr/Val in PLK2 and Gly/Ile in PLK3. In addition, the residues pair Arg516/Phe535 are lysine and tyrosine in PLK2 and PLK3[44], respectively.

Description of Resource Needs: We request high-speed unlimited computation resources (*Maximize ACCESS allocations*) to simulate our protein-inhibitors complexes fast, as we hope to pursue the MetaD method to improve the study scope. Access to a computing resource with GPU-enabled AMBER software will allow us to achieve the research aim. The intended software program for our MetaD simulations is AMBER, with the flexibility to opt for NAMD as a backup. The requested unlimited supercomputing time provided through NSF ACCESS is crucial to executing this research work successfully. **Table 2** summarizes the requested resources for the MetaD simulations of the PLK proteins coupled with approximately 150 molecules on Anvil GPU (www.rcac.purdue.edu/anvil) based on previous simulations with AMBER20. Anvil Purdue infrastructure GPU nodes have 4 GPUs per node, all shared by users. Standard GPU jobs can run up to four days using all 16 full nodes. 1 SU corresponds to using one GPU for one hour, and our benchmark with PLK1-PBD cMD simulations with known inhibitors runs at approximately 18 GPU hours per 200 ns production for one GPU on this infrastructure (with the flexibility to submit six jobs in a row). Based on the exploration for binding at various protein cavities (**Figure 1**) and at least 200 ns production run per protein-ligand replicate, the total (tentative) Anvil GPU hours required for our study is 204436.8 (**Table 2**).

Table 2. Overview of the MetaD simulation details and tentative resources required to simulate the PBD of PLK1–3 per 200 ns simulation length for each replicate.

System	Approximate system size	Hours: mins on 1 GPU	Number of simulations	Total simulation	Total GPU hours: min
PLK1–(–)–9*	70500	18:08	5	1 μs	90.5
PLK2–(–)–9*	70500	18:08	5	1 μs	90.5
PLK3–(–)–9*	70500	18:08	5	1 μs	90.5
PLK1–Allopole-A	74800	18:02	5	1 μs	90.1
PLK2–Allopole-A	74800	18:02	5	1 μs	90.1
PLK3–Allopole-A	74800	18:02	5	1 μs	90.1
PLK1–Poloxin	70800	18:00	5	1 μs	90
PLK2–Poloxin	70800	18:00	5	1 μs	90
PLK3–Poloxin	70800	18:00	5	1 μs	90
PLK1+46 compounds	70900	18:10	25	5 μs × 46	20815
PLK2+46 compounds	70900	18:10	25	5 μs × 46	20815
PLK3+46 compounds	70900	18:10	25	5 μs × 46	20815
PLK1+104 compounds	70900	18:10	25	5 μs × 104	47060
PLK2+104 compounds	70900	18:10	25	5 μs × 104	47060
PLK3+104 compounds	70900	18:10	25	5 μs × 104	47060
Total					**204436.8**

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References

1. Chen, Y. *et al.* Identification of a novel Polo-like kinase 1 inhibitor that specifically blocks the functions of Polo-Box domain. *Oncotarget* **8**, 1234 (2017).
2. Wang, B. *et al.* PLK1 inhibition sensitizes breast cancer cells to radiation via suppressing autophagy. *Int. J. Radiat. Oncol. Biol. Phys.* **110**, 1234–1247 (2021).
3. Su, S., Chhabra, G., Singh, C. K., Ndiaye, M. A. & Ahmad, N. PLK1 inhibition-based combination therapies for cancer management. *Transl. Oncol.* **16**, 101332 (2022).
4. Strebhardt, K. & Ullrich, A. Targeting polo-like kinase 1 for cancer therapy. *Nat. Rev. Cancer* **6**, 321–330 (2006).
5. Knecht, R., Oberhauser, C. & Strebhardt, K. PLK (polo-like kinase), a new prognostic marker for oropharyngeal carcinomas. *Int. J. Cancer* **89**, 535–536 (2000).
6. McInnes, C. & Wyatt, M. D. PLK1 as an oncology target: current status and future potential. *Drug Discov. Today* **16**, 619–625 (2011).
7. Kumar, S. & Kim, J. PLK-1 Targeted Inhibitors and Their Potential against Tumorigenesis. *BioMed Res. Int.* **2015**, 705745 (2015).
8. Eckerdt, F., Yuan, J. & Strebhardt, K. Polo-like kinases and oncogenesis. *Oncogene* **24**, 267–276 (2005).
9. Stratmann, J. A. & Sebastian, M. Polo-like kinase 1 inhibition in NSCLC: mechanism of action and emerging predictive biomarkers. *Lung Cancer Targets Ther.* **10**, 67 (2019).
10. Montaudon, E. *et al.* PLK1 inhibition exhibits strong anti-tumoral activity in CCND1-driven breast cancer metastases with acquired palbociclib resistance. *Nat. Commun.* **11**, 1–16 (2020).
11. Jeong, S. B. & Kang, K. W. Involvement of Polo-like kinase 1 (Plk1) in tumor growth and epithelial-mesenchymal transition of tamoxifen-resistant breast cancer. *FASEB J.* **31**, lb536–lb536 (2017).
12. Jeong, S. B. *et al.* Essential role of polo-like kinase 1 (Plk1) oncogene in tumor growth and metastasis of tamoxifen-resistant breast cancer. *Mol. Cancer Ther.* **17**, 825–837 (2018).
13. Hu, K., Law, J. H., Fotovati, A. & Dunn, S. E. Small interfering RNA library screen identified polo-like kinase-1 (PLK1) as a potential therapeutic target for breast cancer that uniquely eliminates tumor-initiating cells. *Breast Cancer Res.* **14**, 1–15 (2012).
14. Zhang, Z. *et al.* Recent progress in agents targeting polo-like kinases: Promising therapeutic strategies. *Eur. J. Med. Chem.* **217**, 113314 (2021).
15. Soung, N.-K. *et al.* Plk1-dependent and-independent roles of an ODF2 splice variant, hCenexin1, at the centrosome of somatic cells. *Dev. Cell* **16**, 539–550 (2009).
16. Lens, S., Voest, E. E. & Medema, R. H. Shared and separate functions of polo-like kinases and aurora kinases in cancer. *Nat. Rev. Cancer* **10**, 825–841 (2010).
17. Strebhardt, K. Multifaceted polo-like kinases: drug targets and antitargets for cancer therapy. *Nat. Rev. Drug Discov.* **9**, 643–660 (2010).
18. Yang, Y. *et al.* Polo-like kinase 3 functions as a tumor suppressor and is a negative regulator of hypoxia-inducible factor-1α under hypoxic conditions. *Cancer Res.* **68**, 4077–4085 (2008).
19. Zitouni, S., Nabais, C., Jana, S. C., Guerrero, A. & Bettencourt-Dias, M. Polo-like kinases: structural variations lead to multiple functions. *Nat. Rev. Mol. Cell Biol.* **15**, 433–452 (2014).
20. Park, J.-E., Hymel, D., Burke Jr, T. R. & Lee, K. S. Current progress and future perspectives in the development of anti-polo-like kinase 1 therapeutic agents. *F1000Research* **6**, (2017).
21. Strebhardt, K., Becker, S. & Matthess, Y. *Thoughts on the Current Assessment of Polo-like Kinase Inhibitor Drug Discovery*. (Taylor & Francis, 2015).
22. Park, J.-E. *et al.* Putting a bit into the polo-box domain of polo-like kinase 1. *J. Anal. Sci. Technol.* **6**, 1–8 (2015).

23. Lee, K. S., Burke Jr, T. R., Park, J.-E., Bang, J. K. & Lee, E. Recent advances and new strategies in targeting Plk1 for anticancer therapy. *Trends Pharmacol. Sci.* **36**, 858–877 (2015).

24. Abdullah, M. & Guruprasad, L. Structural insights into the inhibitor binding and new inhibitor design to Polo-like kinase-1 Polo-box domain using computational studies. *J. Biomol. Struct. Dyn.* (2018).

25. Murugan, R. N. *et al.* Plk1-targeted small molecule inhibitors: molecular basis for their potency and specificity. *Mol. Cells* **32**, 209 (2011).

26. Watanabe, N. & Osada, H. Small molecules that target phosphorylation dependent protein–protein interaction. *Bioorg. Med. Chem.* **24**, 3246–3254 (2016).

27. McInnes, C. *et al.* Targeting subcellular localization through the polo-box domain: non-ATP competitive inhibitors recapitulate a PLK1 phenotype. *Mol. Cancer Ther.* **11**, 1683–1692 (2012).

28. Vanommeslaeghe, K. & MacKerell Jr, A. D. Automation of the CHARMM General Force Field (CGenFF) I: bond perception and atom typing. *J. Chem. Inf. Model.* **52**, 3144–3154 (2012).

29. Atomwise – Better medicines faster. https://www.atomwise.com/.

30. Humphrey, W., Dalke, A. & Schulten, K. VMD: visual molecular dynamics. *J. Mol. Graph.* **14**, 33–38 (1996).

31. Hanwell, M. D. *et al.* Avogadro: an advanced semantic chemical editor, visualization, and analysis platform. *J. Cheminformatics* **4**, 17 (2012).

32. Liu, Y. *et al.* CB-Dock2: Improved protein–ligand blind docking by integrating cavity detection, docking and homologous template fitting. *Nucleic Acids Res.* **50**, W159–W164 (2022).

33. Yang, X., Liu, Y., Gan, J., Xiao, Z.-X. & Cao, Y. FitDock: Protein–ligand docking by template fitting. *Brief. Bioinform.* **23**, bbac087 (2022).

34. Daina, A., Michielin, O. & Zoete, V. SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Sci. Rep.* **7**, 1–13 (2017).

35. Phillips, J. C. *et al.* Scalable molecular dynamics with NAMD. *J. Comput. Chem.* **26**, 1781–1802 (2005).

36. Ferruz, N., Schmidt, S. & Höcker, B. ProteinTools: a toolkit to analyze protein structures. *Nucleic Acids Res.* **49**, W559–W566 (2021).

37. Xue, L. C., Rodrigues, J. P., Kastritis, P. L., Bonvin, A. M. & Vangone, A. PRODIGY: a web server for predicting the binding affinity of protein–protein complexes. *Bioinformatics* **32**, 3676–3678 (2016).

38. Pires, D. E. & Ascher, D. B. CSM-lig: a web server for assessing and comparing protein–small molecule affinities. *Nucleic Acids Res.* **44**, W557–W561 (2016).

39. Lawal, M. M. & Kucukkal, T. G. Exploration of Binding Diversity and Allosteric Insights into PLK1 Polo-Box Domain in Anticancer Drug Discovery. *J. Phys. Chem. B* **Under Review**, (2024).

40. Lawal, M. M., Fernsebner, S., Lundberg, D. & Kucukkal, T. G. Student Research Project in Medicinal Chemistry: In-silico Study of the Physicochemical, Drug-Likeness, and ADMET-S Properties of Small Molecules. *J. Chem. Educ.* **Under Review**,.

41. Rose, P. W. *et al.* The RCSB protein data bank: integrative view of protein, gene and 3D structural information. *Nucleic Acids Res.* gkw1000 (2016).

42. Reindl, W., Yuan, J., Krämer, A., Strebhardt, K. & Berg, T. Inhibition of polo-like kinase 1 by blocking polo-box domain-dependent protein-protein interactions. *Chem. Biol.* **15**, 459–466 (2008).

43. Park, J.-E. *et al.* Specific inhibition of an anticancer target, polo-like kinase 1, by allosterically dismantling its mechanism of substrate recognition. *Proceedings of the National Academy of Sciences* vol. 120 e2305037120 (2023).

44. Huggins, D. J. *et al.* Development of a Novel Cell-Permeable Protein–Protein Interaction Inhibitor for the Polo-box Domain of Polo-like Kinase 1. *ACS Omega* **5**, 822–831 (2019).

45. Scharow, A. *et al.* Optimized Plk1 PBD inhibitors based on poloxin induce mitotic arrest and apoptosis in tumor cells. *ACS Chem. Biol.* **10**, 2570–2579 (2015).

46. Ensing, B., De Vivo, M., Liu, Z., Moore, P. & Klein, M. L. Metadynamics as a tool for exploring free energy landscapes of chemical reactions. *Acc. Chem. Res.* **39**, 73–81 (2006).

47. Du, X. *et al.* Insights into protein–ligand interactions: mechanisms, models, and methods. *Int. J. Mol. Sci.* **17**, 144 (2016).

48. Laio, A. & Parrinello, M. Escaping free-energy minima. *Proc. Natl. Acad. Sci.* **99**, 12562–12566 (2002).

49. Barducci, A., Bonomi, M. & Parrinello, M. Metadynamics. *Wiley Interdiscip. Rev. Comput. Mol. Sci.* **1**, 826–843 (2011).

50. Branduardi, D., Bussi, G. & Parrinello, M. Metadynamics with adaptive Gaussians. *J. Chem. Theory Comput.* **8**, 2247–2254 (2012).

51. Schäfer, T. M. & Settanni, G. Data Reweighting in Metadynamics Simulations. *J. Chem. Theory Comput.* **16**, 2042–2052 (2020).

52. Šućur, Z. & Spiwok, V. Sampling Enhancement and Free Energy Prediction by the Flying Gaussian Method. *J. Chem. Theory Comput.* **12**, 4644–4650 (2016).

53. Sun, R., Dama, J. F., Tan, J. S., Rose, J. P. & Voth, G. A. Transition-tempered metadynamics is a promising tool for studying the permeation of drug-like molecules through membranes. *J. Chem. Theory Comput.* **12**, 5157–5169 (2016).

54. Barducci, A., Bussi, G. & Parrinello, M. Well-tempered metadynamics: a smoothly converging and tunable free-energy method. *Phys. Rev. Lett.* **100**, 020603 (2008).

55. Valsson, O., Tiwary, P. & Parrinello, M. Enhancing important fluctuations: Rare events and metadynamics from a conceptual viewpoint. *Annu. Rev. Phys. Chem.* **67**, 159–184 (2016).

56. Yang, L., Liu, C.-W., Shao, Q., Zhang, J. & Gao, Y. Q. From thermodynamics to kinetics: Enhanced sampling of rare events. *Acc. Chem. Res.* **48**, 947–955 (2015).

57. Bussi, G. & Branduardi, D. Free-energy calculations with metadynamics: Theory and practice. *Rev. Comput. Chem. Vol. 28* 1–49 (2015).

58. Salvalaglio, M., Tiwary, P. & Parrinello, M. Assessing the reliability of the dynamics reconstructed from metadynamics. *J. Chem. Theory Comput.* **10**, 1420–1425 (2014).

59. Dickson, A., Tiwary, P. & Vashisth, H. Kinetics of ligand binding through advanced computational approaches: a review. *Curr. Top. Med. Chem.* **17**, 2626–2641 (2017).

60. Raiteri, P., Laio, A., Gervasio, F. L., Micheletti, C. & Parrinello, M. Efficient reconstruction of complex free energy landscapes by multiple walkers metadynamics. *J. Phys. Chem. B* **110**, 3533–3539 (2006).

61. Bonomi, M. *et al.* PLUMED: A portable plugin for free-energy calculations with molecular dynamics. *Comput. Phys. Commun.* **180**, 1961–1972 (2009).

62. Case, D. A. *et al. Amber 2021*. (University of California, San Francisco, 2021).

63. Van Der Spoel, D. *et al.* GROMACS: fast, flexible, and free. *J. Comput. Chem.* **26**, 1701–1718 (2005).

64. Yuan, J. *et al.* Polo-box domain inhibitor poloxin activates the spindle assembly checkpoint and inhibits tumor growth in vivo. *Am. J. Pathol.* **179**, 2091–2099 (2011).

65. Yin, Z., Song, Y. & Rehse, P. H. Thymoquinone Blocks pSer/pThr Recognition by Plk1 Polo-Box Domain As a Phosphate Mimic. *ACS Chem. Biol.* **8**, 303–308 (2013).

66. Archambault, V. & Normandin, K. Several inhibitors of the Plk1 Polo-Box Domain turn out to be non-specific protein alkylators. *Cell Cycle* **16**, 1220–1224 (2017).

67. Reindl, W., Yuan, J., Krämer, A., Strebhardt, K. & Berg, T. A pan-specific inhibitor of the polo-box domains of polo-like kinases arrests cancer cells in mitosis. *Chembiochem* **10**, 1145–1148 (2009).

68. Watanabe, N. *et al.* Deficiency in chromosome congression by the inhibition of Plk1 polo box domain-dependent recognition. *J. Biol. Chem.* **284**, 2344–2353 (2009).

69. Shan, H.-M., Shi, Y. & Quan, J. Identification of Green Tea Catechins as Potent Inhibitors of the Polo-Box Domain of Polo-Like kinase 1. *ChemMedChem* **10**, 158–163 (2015).

70. Cheng, K.-Y., Lowe, E. D., Sinclair, J., Nigg, E. A. & Johnson, L. N. The crystal structure of the human polo-like kinase-1 polo box domain and its phospho-peptide complex. *EMBO J.* **22**, 5757–5768 (2003).

71. Normandin, K. *et al.* Identification of Polo-like kinase 1 interaction inhibitors using a novel cell-based assay. *Sci. Rep.* **6**, 37581 (2016).

72. Sharma, P. *et al.* A cryptic hydrophobic pocket in the polo-box domain of the polo-like kinase PLK1 regulates substrate recognition and mitotic chromosome segregation. *Sci. Rep.* **9**, 15930 (2019).

73. Lindenburg, L. H. *et al.* Improved RAD51 binders through motif shuffling based on the modularity of BRC repeats. *Proc. Natl. Acad. Sci.* **118**, e2017708118 (2021).

74. Śledź, P. *et al.* From Crystal Packing to Molecular Recognition: Prediction and Discovery of a Binding Site on the Surface of Polo-Like Kinase 1. *Angew. Chem. Int. Ed.* **50**, 4003–4006 (2011).

75. Iannuzzi, M., Laio, A. & Parrinello, M. Efficient exploration of reactive potential energy surfaces using Car-Parrinello molecular dynamics. *Phys. Rev. Lett.* **90**, 238302 (2003).

76. Babin, V., Roland, C. & Sagui, C. Adaptively biased molecular dynamics for free energy calculations. *J. Chem. Phys.* **128**, (2008).

77. Moradi, M., Babin, V., Roland, C. & Sagui, C. The Adaptively Biased Molecular Dynamics method revisited: New capabilities and an application. in *Journal of Physics: Conference Series* vol. 640 012020 (IOP Publishing, 2015).

78. Lauck, F., Smith, C. A., Friedland, G. F., Humphris, E. L. & Kortemme, T. RosettaBackrub—a web server for flexible backbone protein structure modeling and design. *Nucleic Acids Res.* **38**, W569–W575 (2010).

79. Tian, C. *et al.* ff19SB: Amino-acid-specific protein backbone parameters trained against quantum mechanics energy surfaces in solution. *J. Chem. Theory Comput.* **16**, 528–552 (2019).

80. Jakalian, A., Bush, B. L., Jack, D. B. & Bayly, C. I. Fast, efficient generation of high-quality atomic charges. AM1-BCC model: I. Method. *J. Comput. Chem.* **21**, 132–146 (2000).

81. Salomon-Ferrer, R., Gotz, A. W., Poole, D., Le Grand, S. & Walker, R. C. Routine microsecond molecular dynamics simulations with AMBER on GPUs. 2. Explicit solvent particle mesh Ewald. *J. Chem. Theory Comput.* **9**, 3878–3888 (2013).

82. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).

83. Waterhouse, A. *et al.* SWISS-MODEL: homology modelling of protein structures and complexes. *Nucleic Acids Res.* **46**, W296–W303 (2018).

84. Sorokina, M., Merseburger, P., Rajan, K., Yirik, M. A. & Steinbeck, C. COCONUT online: collection of open natural products database. *J. Cheminformatics* **13**, 1–13 (2021).